WEEK 3 · DAY 3 · INSTRUCTOR NOTES

The Slow Creep: Drift and CUSUM

For the trainer only.

The one idea for today: *some faults never produce a single scary reading. You catch them by adding up small leftovers (CUSUM) or by watching the speed of change (rolling slope).*

At a glance

Length	~2 hours, including 15 minutes for the final-project Checkpoint 2
Participants leave able to	name spike / level shift / drift; write CUSUM from scratch; set k and h from σ; tell alarm date from start date; watch a rolling slope for drift
Covers (original)	Week 4 topic *change points and drift*; Week 4 Code Book sections **2. TWTA thermal** and **3. Battery health**
Datasets	`Time-series/week-04/data/twta_thermal_daily.csv`, `Time-series/week-04/data/battery_health_daily.csv` (the **week-04** versions, the notebook default)
Files you need open	animated slides, your **solution** notebook, the rule book PDF, the final-project checkpoint table

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Hook**	Landlord's rent: small-small increases add up to ₦415k	1–3
0:08	**Three shapes**	spike vs level shift vs drift	4
0:12	**The amplifier**	3σ rule fires 0 times; zoom: 6.9σ locally; take away what you know	5–8
0:25	**CUSUM**	the one line; by-hand table (press → to fill S); k and h	9–11
0:35	**CUSUM in action**	real step caught on day 1; small practice fault caught in 3 days; choosing h	12–14
0:45	**The battery**	never crosses 95 Ah; speed 1.6× faster; rolling slope crosses −3.0	15–18
0:50	**Toolbox**	right tool for each shape	19
0:52	**Hands-on**	Part A (15 min), Part B (30 min), Part C (15 min)	21
1:45	**Checkpoint 2**	one by one: unit head's reply + profiled data (see below)	20

Time	Block	What happens	Slides
1:55	**Wrap-up**	recap; tease the answer key on Week 4 Day 1	22

Explain it like this

- **Spike / level shift / drift.** A spike goes up and comes back. A level shift steps up and stays. A drift leans: the slope changes.
- **Why a 3σ rule misses the amplifier's step.** σ of the whole two years (1.97 °C) is inflated by the yearly sun-angle swing and the rising drive level, so the ceiling sits at 69.81 °C. The hottest day ever is 68.15 °C, so it fires 0 times. Against the 30 days before (σ 0.32 °C), the +2.17 °C jump is 6.9σ.
- **Take away what you already know.** `expected_temp()` learns "normal" from 2024: drive level + sun angle (sin/cos of day of year) + slow ageing (a straight line in time). Its second pass ignores the 2-day hot spike of 3–4 Sep 2024 so it doesn't bend "normal". The leftovers have σ = 0.20 °C, average −0.01 °C from January to 11 May 2025, and +1.83 °C from 12 May. That matches the +1.8 °C step hidden in the file.
- **CUSUM.** Every day: `S = max(0, S + leftover − k)`. Small positive leftovers pile up; the slack k stops ordinary noise piling up; the floor at zero stops calm days banking credit. Alarm when S > h.
- **k and h.** Both are multiples of σ of the leftovers in a calm period: k = 0.5σ (0.10 °C), h = 5σ (1.02 °C).
- **Alarm date ≠ start date.** CUSUM needs evidence, so it always rings after the change began. Best guess of the start: the day after S was last zero.
- **Rolling slope.** The battery never crosses 95 Ah, so a threshold sleeps. The fade speed goes from −2.24 to −3.68 Ah per year. A 90-day trailing slope crosses a −3.0 warning line 31 days after the true change, because a trailing window needs time to fill with the new behaviour.

Slide talking points

#	Say this	Ask this
3	"No single year shocks you. The total does."	"Who has a landlord like this?" 😄
6	"Zero alarms in two years. Is the amplifier fine?"	"Can you spot the step?"
7	65.20 → 67.37 °C. 6.9σ locally, 1.1σ against the whole series.	"Why does the same jump look big and small?"
10	Press → seven times; S fills in. Day 4 starts the climb; day 7 alarms.	"When did it START?" (day 4)
12	S was 0.90 on 11 May, 2.86 on 12 May; h = 1.02. Same-day alarm because the step is ~9σ of leftovers.	–
13	The point rule saw 0 hits in 30 days. CUSUM: 13 Feb. Trace back: 10 Feb. ✅	"How did we know it was 10 Feb?"
14	2σ: 10 false alarms / 1 day. 5σ: 0 / 3 days. 10σ: 0 / 6 days.	"Which would you choose for an amplifier? For an email alert to the MD?"
18	31 days late, and that is the price of a trailing window.	–

Answer key

Task	Answer	Common mistake → what to say
A1	`"spike"`, `"level shift"`, `"drift"`	"level_shift" or "Level Shift" are accepted
A2	`S = max(0, S + x − k)` → `[0, 0, 0, 0.5, 1.2, 1.6, 2.2]`	forgetting `max(0, …)` → negative values; forgetting `− k` → noise piles up
A3	start day **4**, alarm day **7**	answering 3 (the last zero), not the day after
A4	`"b"` (slack)	"a": k is not the alarm line; that's h
B1	**0** days above 69.81 °C	–
B2	**6.9** σ (65.20 → 67.37 °C, local sd 0.32)	dividing by the whole-series sd gives 1.1
B3	**+1.83** °C	forgetting to fill `expected`
B4	k = **0.102**, h = **1.016** °C (σ = 0.2031)	swapping k and h
B5	**2025-05-12** (delay 0 days)	–
B6	alarm **2025-02-13**, start **2025-02-10**	giving `last_zero` (9 Feb) instead of the day after
C1	**0** days below 95 Ah (lowest 95.39)	using the week-03 file
C2	**−2.24** → **−3.68** Ah/yr (1.6× faster)	slicing `"2025-04-01"` into both halves
C3	**2025-05-02** (31 days after 1 Apr 2025)	typing `3.0` instead of `-3.0`
🌶️	**10** false alarms with h = 2σ in 404 calm days	counting all alarms, not just those before the start

The secret fault (instructor only): `practice_residuals()` adds **+0.3 °C (≈1.5σ) from 10 Feb 2025** to the amplifier's calm leftovers (Jan 2024 to 11 May 2025, minus the 2 spike days). A 3σ point rule catches 0 of the first 30 days.

Quiz link

Original Week 4 quiz (`Time-series/MCQ/week-04-mcq.csv`), topic **"change points and drift"**, questions **19-24**:

Q	Taught on
19 Shift vs drift	slide 4, task A1
20 What CUSUM accumulates	slide 9, task A2
21 The slack k	slide 11, tasks A4, B4
22 Alarm date vs onset (False)	slides 10, 13, task A3, B6
23 Battery fade knee → rolling slope	slides 17–18, task C3

Q	Taught on
24 TWTA 6.9σ missed by a 3σ point rule	slides 6–7, tasks B1–B2

These come up again in Quiz 2 on Week 4 Day 1.

🏁 Final project · Checkpoint 2: 📩 Data in hand (15 min)

Matches `week-6/final-project/Final-Project-Requirements.pdf`. Go round one by one:

Check	Pass if
Unit head's agreement	an **email reply** naming the process, the data and who it's for
Data in hand	loaded with a DatetimeIndex, one time zone, `.asfreq()`
Profile	rows, date range, gaps, duplicated timestamps reported
Honest plot	the operational threshold drawn, with a title that states a finding
Data rules	real data on NIGCOMSAT systems only (not Colab); nothing raw on GitHub

No reply yet? Agree a follow-up date with them today. Anyone planning to use only synthetic data needs **written** agreement from their unit head that no real data can be made available. Next checkpoint: **Week 4 · Day 3: method drafted.**

Troubleshooting

Problem	Fix
CUSUM fires in January	`- k` is missing in the loop, or k and h were swapped
B6 start one day early	they returned `last_zero`; it must be `last_zero + pd.Timedelta(days=1)`
Battery numbers don't match	they loaded `week="week-03"`; use the default (week-04)
`rolling_slope` feels slow	it fits 1,000+ small lines; a few seconds is normal

Notes on the original material

- The original Week 4 code book has **no CUSUM code**; we wrote it from scratch.
- Original battery fade rates: *"−2.4 Ah/year before, −4.0 after"*. The data gives **−2.24 and −3.68** (a straight-line fit before and after 1 April 2025). The slides use the data's numbers.
- The original says the TWTA step is *"6.9σ against local spread"*: correct (65.20 → 67.37 °C, local sd 0.32).
- The **week-03** and **week-04** files named `battery_health_daily.csv` and `twta_thermal_daily.csv` differ: only week-04 has the step, the 2-day hot spike and the fade speed-up. The notebook default is week-04.